Dispersed phase Mixing

Coalescence & breakup occurs

Cell Mass Model

$$\frac{\partial n(m,t)}{\partial t} + \frac{\partial}{\partial m}[\dot{M} n] = \underbrace{-I(m,s)\, n(m,t)}_{\text{Change in no density (due to division)}} + 2\int_m^\infty I(m',s)\, p(m|m',s)\, n(m',t) \quad \because \int_0^{m'} p(m|m',s)\, dm = 1$$

These two events are the same $\rightarrow p(m|m',s) = p(m'-m|m',s)$

$$\int_0^{m'} m\, p(m|m',s)\, dm = \frac{1}{2} m'$$

Cell division does not change overall mass

Total no. of cells $\rightarrow N(t) = \int_0^\infty n(m,t)\, dm$

$$\Rightarrow \frac{dN}{dt} + \left[\dot{M}(m,s)\, n(m,t)\, dm\right]_0^\infty = -\int_0^\infty I(m,s)\, n(m,t)\, dm + 2\int_0^\infty dm \int_m^\infty dm'\, I(m',s)\, p(m|m',s)\, n(m',t)\, dm'$$

$$\frac{dN}{dt} = -\int_0^\infty p(m,s)\, n(m,t)\, dm + 2\int_0^\infty dm' \int_0^{m'} dm\, I(m',s)\, n(m',t)\, p(m|m',s) = \int_0^\infty p(m,s)\, n(m,t)\, dm$$

$$\frac{\partial n(x,t)}{\partial t} = \frac{1}{2}\int_0^x a(x-y,y)\, n(x-y,t)\, n(y,t)\, dy - \int_0^\infty a(x,y)\, n(x,t)\, n(y,t)\, dy$$

$$\frac{dN}{dt} = \frac{1}{2}\int_0^\infty dx \int a(x-y,y)\, n(x-y,t)\, n(y,t)\, dy - \int_0^\infty dx \int_0^\infty dy\, a(x,y)\, n(x,t)\, n(y,t)$$

$$= \frac{1}{2}\int_0^\infty dy \int a(x-y,y)\, n(x-y,t)\, n(y,t)\, dx \qquad x-y = u,\ dx = du$$

$$= \frac{1}{2}\int_0^\infty dy \int_0^\infty dx\, a(x,y)\, n(y,t)\, n(x,t)\, dy$$

$n(\underset{\sim}{x}, \underset{\sim}{r}, t) \Rightarrow \frac{\partial n}{\partial t} + \nabla_{\underset{\sim}{x}} \cdot \dot{\underset{\sim}{x}}(\underset{\sim}{x},\underset{\sim}{r}) + \nabla_{\underset{\sim}{r}} \cdot \dot{\underset{\sim}{r}}\, n$

Monte Carlo simulations: if you know the distribution → You can generate random variables to model that distribution

Precipitation Reaction:

occurs because system wants to reach equilibrium

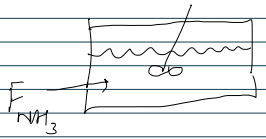
→ Solubility products is exceeded (leads to supersaturation)

→ Mixing of two liquids giving rise to chemical reactions b/w ionic species

} You have to play w/

1D model, L, t ← How long is the crystal grown

L: characteristic length



$$NH_3^{(aq)} \overset{K_b}{\rightleftharpoons} NH_4^+ + OH^-$$

$$K_b = \frac{[NH_4^+][OH^-]}{[NH_3^{aq}]}$$

$$M^{a+} + \alpha OH^- \rightarrow \underbrace{M(OH)_\alpha}_{\text{precipitation reaction}}$$

$$K_s = [M^{a+*}][OH^{-*}]^\alpha \qquad * \text{ denotes equilibrium}$$

$K_s = [\quad][\quad]^*$ denotes equilibrium

↳ solubility product

supersaturation

↳ $\Lambda = \frac{[M]}{C^2}$

$K_w = 10^{-14} = [H^+][OH^-] = [H^+]C = K_w$

↑
ionic product of water

Principle of electroneutrality: Anion A^- remains at its original level of $[M]_0$ (since it does not participate in the reaction)

↑ precipitate particles are consuming OH^-

↑ Have to donate OH^- to maintain K_w

L → rate of growth of particle

M → precipitate